

Environment / Stream

Event stream
 (u_t, v_t^*, t)

Performative Feedback

Update backbone
& graph stats

Policy + Logging

Candidate Set
 $C_t = \{v_t^*\} \cup N_t$

Backbone
Scoring
 $f_t \rightarrow \hat{p}_t$

Decision-
layer logit
shift

Apply
 $r^{(0)}$ -calibrator

Coverage
 $\rightarrow$ modifier+ mask

Policy π_t
(topK-Stoch/
 ε -greedy)

Logged
propensities
 $\hat{e}_t$

Observe
outcome y_t

Deployed
hit?

NO

No update
(bandit)

Estimation

(Cross-fit + GA-DR)

DR ingest

Online cross-fit
 $\hat{\mu}_0, \hat{\mu}_1$

GA-DR
buffer

Auditing + Control

Update
Multicalibrator
(audit checkpoints)

Fairness metrics
 $g_{gap}^{TE}, g_{max}^{Cal}$

Residual-OI auditor
Residual-OI bounds

PI dual update
 $\lambda_{TE}, \lambda_{Cal}, \lambda_{Min}$

